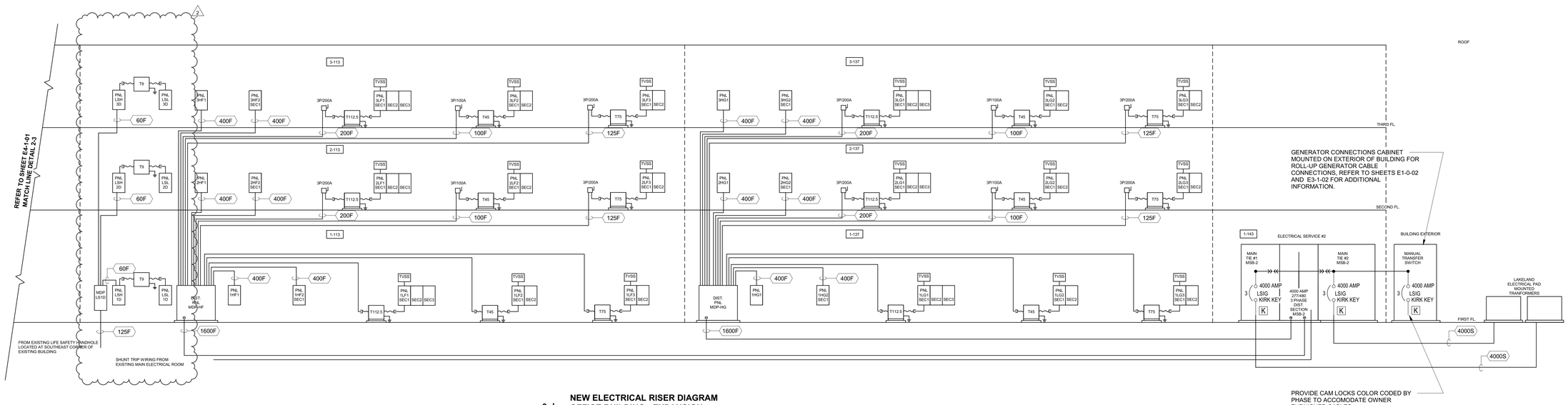
Building Inspection Division
THESE PLANS HAVE BEEN REVIEWED FOR COMPLIANCE WITH THE CURRENT EDITION OF THE FLORIDA BUILDING CODE
Date: 11/15/19
Permit Number: BLD19-06035



3 | NEW ELECTRICAL RISER DIAGRAM
E4-1-02 | OFFICE BUILDING - EXPANSION

Scale: N.T.S.

FEEDER SCHEDULE

TAG	PARALLEL RUNS	CONDUCTORS (75 DEG C)	CONDUIT TRADE SIZE
60F	N/A	(4) 6 AWG & (1) 10 AWG G	1"
100F	N/A	(4) 1 AWG & (1) 8 AWG G	1-1/2"
125F	N/A	(4) 1 AWG & (1) 6 AWG G	1-1/2"
150F	N/A	(4) 1/0 AWG & (1) 6 AWG G	1-1/2"
200F	N/A	(4) 3/0 AWGI & (1) 6 AWG G	2"
225F	N/A	(4) 4/0 AWG & (1) 4 AWG G	2-1/2"
400F	2 PARALLEL RUNS	(4) 3/0 AWG & (1) 3 AWG G	2 1/2"
500F	2 PARALLEL RUNS	(4) 250 kcmil & (1) 2 AWG G	2-1/2"
600F	2 PARALLEL RUNS	(4) 350 kcmil & (1) 1 AWG G	3"
800F	2 PARALLEL RUNS	(4) 500 kcmil & (1) 1/0 AWG G	4"
1200F	3 PARALLEL RUNS	(4) 600 kcmil & (1) 3/0 AWG G	4"
1600F	4 PARALLEL RUNS	(4) 600 kcmil & (1) 4/0 AWG G	4"
2000F	5 PARALLEL RUNS	(4) 600 kcmil & (1) 250 kcmil G	4"

GENERAL NOTES:
1. ALL CONDUCTORS LISTED IN THIS SCHEDULE ARE COPPER, UNLESS OTHERWISE NOTED.
2. WHERE PARALLEL RUNS ARE INDICATED, PROVIDE THE CONDUCTOR QUANTITY AND CONDUIT SIZE FOR EACH PARALLEL RUN.

SERVICE LATERAL SCHEDULE

TAG	PARALLEL RUNS	CONDUCTORS (75 DEG C)	CONDUIT TRADE SIZE
4000S	10 PARALLEL RUNS	4-600 kcmil COPPER	4"

GENERAL NOTES:
1. WHERE PARALLEL RUNS ARE INDICATED, PROVIDE THE CONDUCTOR QUANTITY AND CONDUIT SIZE FOR EACH PARALLEL RUN.

FAULT CURENT SCHEDULE

PANEL/EQUIPMENT NAME	FED FROM	AVAILABLE FAULT CURRENT
MDP LS1D	EXISTING PANEL EM	22,762
LSH 1D	MDP LS1D	12,801
LSH 2D	MDP LS1D	9,275
LSH 3D	MDP LS1D	7,272
MSB-2	UTILITY XFMR	29,963
MDP-HG	MSB-2	21,583
1HG1	MDP-HG	18,935
1HG2	MDP-HG	18,482
2HG1	MDP-HG	17,637
2HG1	MDP-HG	17,243
3HG1	MDP-HG	19,912
3HG2	MDP-HG	19,912
MDP-HF	MSB-2	16,866
1HF1	MDP-HF	18,935
1HF2	MDP-HF	18,482
2HF1	MDP-HF	17,637
2HF2	MDP-HF	17,243
3HF1	MDP-HF	19,912
3HF2	MDP-HF	19,912

GENERAL NOTES:
1. AVAILABLE FAULT CURRENT (AFC) IS THE MINIMUM AIC RATING REQUIREMENT OF THE PANEL/CIRCUIT BREAKERS/EQUIPMENT.
2. UTILITY TRANSFORMER IS 1500 KVA, 277/480 VOLT, 3 PHASE, WITH 33,915 AIC RATING AT SECONDARY LUGS.
3. IN ACCORDNACE WITH NEC 110.24 PROVIDE PLAQUE ON ALL MAIN SERVICE DISCONNECTS INDICATING:
   A. "MAIN SERVICE DISCONNECT # OF #".
   B. MAXIMUM AVAILABLE FAULT CURRENT.
   C. FAULT CURRENT CALCULATION DATE OF OCT 15, 2019.

TRANSFORMER SCHEDULE

ITEM	KVA	PRIMARY VOLTS - PHASE	COPPER CONDUCTORS (75°C)	THHN-THWN in FMC CONDUIT TRADE SIZE	SECONDARY VOLTS - PHASE	COPPER CONDUCTORS (75°C)	NEC 250.66 COPPER GRD.ELEC. CONDUCT. BONDING JUMPER	THHN-THWN in FMC CONDUIT TRADE SIZE
T9	9	480 - 3	(3) 12 AWG & (1) 12AWG G	3/4"	120/208 - 3	(4) 10 AWG & (1) 10 AWG G	6 AWG	3/4"
T15	15	480 - 3	(3) 10 AWG & (1) 10 AWG G	3/4"	120/208 - 3	(4) 4 AWG & (1) 10 AWG G	6 AWG	1-1/4"
T30	30	480 - 3	(3) 6 AWG & (1) 10 AWG G	3/4"	120/208 - 3	(4) 1 AWG & (1) 8 AWG G	6 AWG	1-1/2"
T45	45	480 - 3	(3) 4 AWG & (1) 8 AWG G	1-1/4"	120/208 - 3	(4) 1/0 AWG & (1) 6 AWG G	6 AWG	2"
T75	75	480 - 3	(3) 1 AWG & (1) 6 AWG G	1-1/2"	120/208 - 3	(4) 4/0 AWG & (1) 4 AWG G	2 AWG	2-1/2"
T112.5	112.5	480 - 3	(3) 2/0 AWG & (1) 6 AWG G	2"	120/208 - 3	2 PARALLEL RUNS OF (4) 3/0 AWG & (1) 2 AWG G EACH RUN	3/0 AWG	2-1/2" EACH RUN

GENERAL NOTES:

1. ENGINEER ESTIMATE 52296 AIC AT THE SECONDARY OF MAIN TRANSFORMER. CONTRACTOR SHALL COORDINATE WITH UTILITY COMPANY(LAKELAND ELECTRIC) AND CONFIRM KAIC AT THE SECONDARY OF THE UTILITY TRANSFORMERS TO CONFORM AIC RATING OF MAIN SWITCHGEAR AND ELECTRICAL PANELBOARDS. CONTRACTOR SHALL SUBMIT SHORT CIRCUIT CALCULATIONS AND SHOP DRAWINGS FOR ENGINEER'S APPROVAL.
2. PROVIDED PERMANENT PLAQUE IN ELECTRICAL ROOM INDICATING LOCATION OF FIRE PUMP SERVICE.
3. PROVIDED PERMANENT PLAQUE IN ELECTRICAL ROOM INDICATING TYPE AND LOCATION OF EMERGENCY GENERATOR AND WARNING SHOCK HAZARD EXISTS IF GROUNDING ELECTRODE CONDUCTOR OR BONDING JUMPER CONNECTION IN THIS EQUIPMENT IS REMOVED WHILE ALTERNATE SOURCE IS ENERGIZED. LOCATE PLAQUE AT MSB-2 INDICATING EMERGENCY GENERATOR LOCATED ON WEST END OF PROPERTY IS SUPPLYING EMERGENCY EQUIPMENT INSIDE THE BUILDNG IN ACCORDANCE WITH NEC 700.7(A) AND 700.7(B).
4. PROVIDE PERMANENT PLAQUES IN THE NEW AND EXISTING MAIN ELECTRICAL ROOMS INDICATING LOCATIONS OF MAIN ELECTRICAL SERVICES IN EACH LOCATION. EXISTING MAIN ELECTRICAL SERVICE IS EQUIPPED WITH SHUNT TRIP BUTTONS. NEW ELECTRICAL SERVIC E WILL HAVE SHUNT TRIP CAPABILITIES WITH THE SHUNT TRIP BUTTON LOCATED ADJACENT TO EXISATING. REFER TO SHEET E3-1-01 FOR LOCATION OF NEW AND EXISTING SHUNT TRIP BUTTONS.
5. BOND ANY METAL PIPING SYSTEM LIKELY TO BECOME ENERGIZED, TO GROUNDING SYSTEM.
6. PROVIDE CONCRETE HOUSE KEEPING PAD FOR ALL FLOOR STANDING ELECTRICAL DISTRIBUTION EQUIPMENT AND DRY TYPE TRANSFORMERS.
7. AS PART OF SHOP DRAWING SUBMITTAL, PROVIDE A DIMENSIONED PLAN VIEW LAYOUT OF EACH ELECTRICAL ROOM AND THE PROPOSED GEAR. AT MINIMUM SUBMITTED PLAN VIEW SHALL INDICATE NEC WORKING CLEARANCES AND ITEMS INSTALLED BY OTHER TRADES (I.E. CHILLED WATER LINES, DOORS, FIRE ALARM PANELS).
8. PROVIDE TYPE 1 SPD ON SERVICE ENTRANCE RATED EQUIPMENT AND TYPE 2 SPD IN DOWNSTREAM PANELBOARDS AND DISTRIBUTION BOARDS.
9. PROVIDE GROUND BAR IN ALL ELECTRICAL AND IDF ROOMS. CONNECT GROUND BARS VIA 3/0 COPPER AND TERMINATE AT GROUNDING ELECTRODE SYSTEM CONNECTION. CONNECT NEW GROUNDING ELECTRODE SYSTEM WITH EXISTING STEEL STRUCTURE VIA 3/0 COPPER IN ACCORDANCE WITH NEC 250.58. REFER TO GROUNDING DETAILS ON SHEET E6-1-01.
10. RECEPTACLE CONTROLS SHALL BE CONTACTOR CONTROLLED PANELBOARDS VIA TIMECLOCK. PROVIDE A ASTRONOMICAL 24/7 PRAGRAMMABLE TIMECLOCK IN EACH ELECTRICAL ROOM TO CONTROL CONTACTOR. COORDINATE WITH OWNER NORMAL HOURS OF OPERATION TO SET TIMECLOCK. REFER TO PANEL SCHEDULES FOR PANEL WITH CONTACTORS.
11. CONTRACTOR SHALL OBTAIN SELECTIVE COORDINATION VERIFICATION OF EXISTING PANEL EM, AND THE NEW PANEL MDPLS1D. SELECTIVE COORDINATION SHALL BE BASED ON MANUFATCURER'S DATA FROM SHOP DRAWINGS SUBMISSIONS.
12. FOR PANELS GREATER THAN 1200 AMPS, PROVIDE ARC ENERGY REDUCTION IN ACCORDANCE WITH NEC 240.87.
13. PER SPECIFICATIONS SECTION 260913 - ELECTRICAL POWER MONITORING AND CONTROL AND IN ACCORDANCE WITH ASHRAE 90.1 SECTION 8.4.3 ELECTRICAL ENERGY MONITORING, THE ELECTRICAL CONTRACTOR SHALL PROVIDE BRANCH CIRCUIT POWER MONITORING EQUAL TO POWERLINK MVP SERIES OR APPROVED EQUAL. THE FOLLOWING CATEGORIES SHALL BE MONITORED SEPARATELY TO PROVIDE TOTAL USAGE FOR EACH CATEGORY LISTED:
   A. TOTAL ELECTRICAL ENERGY OF BUILDING (MAIN SWITCHBOARD SWBD1)
   B. HVAC SYSTEMS
   C. INTERIOR LIGHTING
   D. EXTERIOR LIGHTING
   E. RECEPTACLE CIRCUITS

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BATSON-COOK CONSTRUCTION

PROJECT

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ISSUE/REVISION

I/R	DATE	DESCRIPTION
2	OCT 25,2019	BUILDING DEPARTMENT COMMENTS
1	SEP 30, 2019	100% CONSTRUCTION DOCUMENTS
0	AUG 23, 2019	CONSTRUCTION DOCUMENTS

PROJECT NUMBER

60601397

SHEET TITLE

ELECTRICAL POWER RISER (CONT.)

SHEET NUMBER

E4-1-02

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